



Quantitative chemistry moles and yield

Name:

Class:

Date:

Time:

63 minutes

Marks:

60 marks

Comments:

1.

This question is about the halogens.

Table 1 shows the melting points and boiling points of some halogens.

Table 1

Element	Melting point in °C	Boiling point in °C
Fluorine	−220	−188
Chlorine	−101	−35
Bromine	−7	59

(a) What is the state of bromine at 0 °C **and** at 100 °C?

Tick (✓) **one** box.

State at 0 °C

State at 100 °C

Gas

Gas

☐

Gas

Liquid

☐

Liquid

Gas

☐

Liquid

Liquid

☐

Solid

Gas

☐

Solid

Liquid

☐

(1)

(b) Explain the trend in boiling points of the halogens shown in **Table 1**.

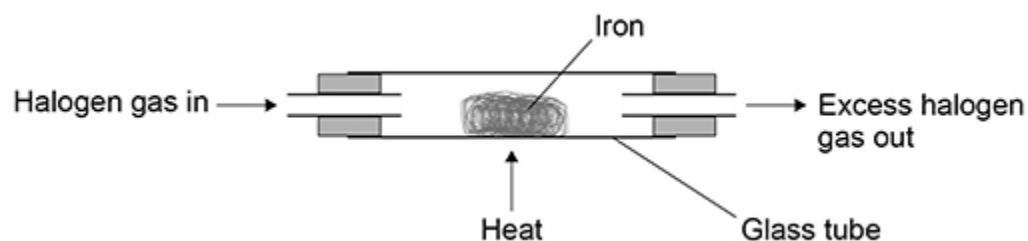
(4)

(c) Why is it **not** correct to say that the boiling point of a single bromine molecule is 59 °C?

(1)

Iron reacts with each of the halogens in their gaseous form.

The diagram below shows the apparatus used.



(d) Give **one** reason why this experiment should be done in a fume cupboard.

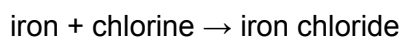
(1)

- (e) Explain why the reactivity of the halogens decreases going down the group.

(3)

- (f) A teacher investigated the reaction of iron with chlorine using the apparatus in the above diagram.

The word equation for the reaction is:



The teacher weighed:

- the glass tube
- the glass tube and iron before the reaction
- the glass tube and iron chloride after the reaction.

Table 2 shows the teacher's results.

Table 2

	Mass in g
Glass tube	51.56
Glass tube and iron	56.04
Glass tube and iron chloride	64.56

Calculate the simplest whole number ratio of:

moles of iron atoms : moles of chlorine atoms

Determine the balanced equation for the reaction.

Relative atomic masses (A_r): Cl = 35.5 Fe = 56

Moles of iron atoms : moles of chlorine atoms = _____ : _____

Equation for the reaction _____

(6)

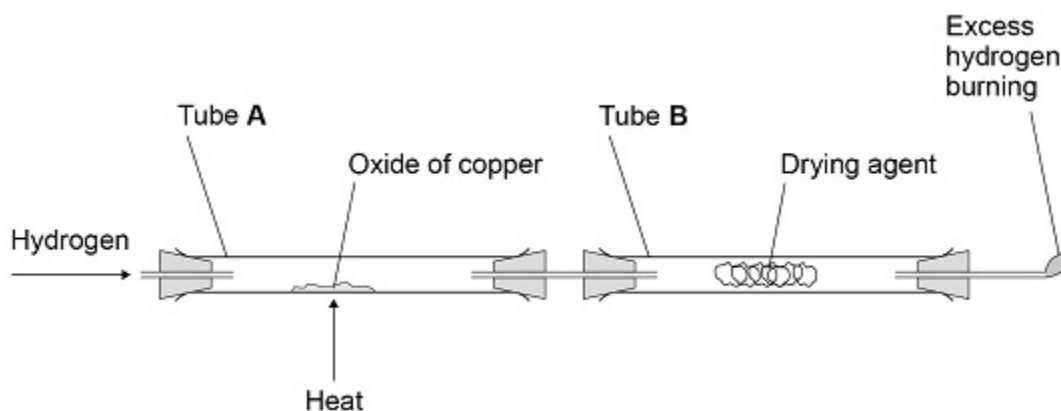
(Total 16 marks)

2.

Copper forms two oxides, Cu_2O and CuO

A teacher investigated an oxide of copper.

The following figure shows the apparatus.



This is the method used.

1. Weigh empty tube **A**.
2. Add some of the oxide of copper to tube **A**.
3. Weigh tube **A** and the oxide of copper.
4. Weigh tube **B** and drying agent.
5. Pass hydrogen through the apparatus and light the flame at the end.
6. Heat tube **A** for 2 minutes.
7. Reweigh tube **A** and contents.
8. Repeat steps 5 to 7 until the mass no longer changes.
9. Reweigh tube **B** and contents.
10. Repeat steps 1 to 9 with different masses of the oxide of copper.

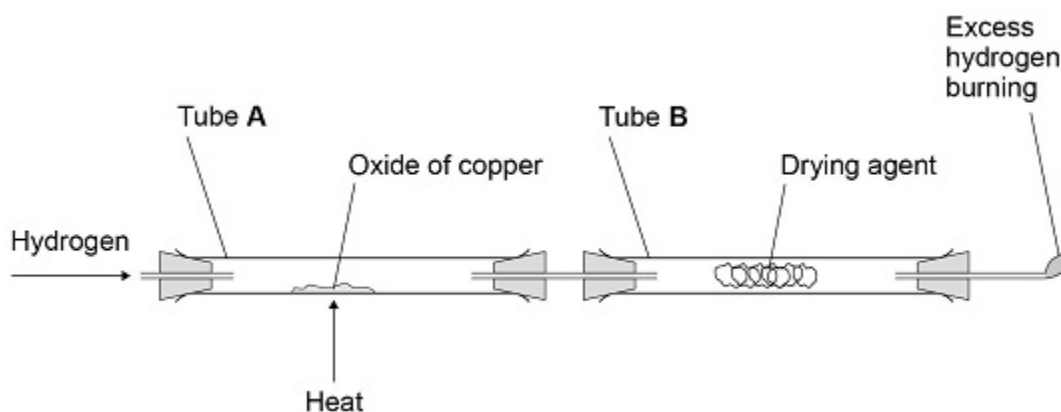
(a) Suggest **one** reason why step 8 is needed.

(1)

(b) Explain why the excess hydrogen must be burned off.

(2)

The figure above is repeated here.



The table below shows the teacher's results.

	Mass in g
Tube A empty	105.72
Tube A and oxide of copper before heating	115.47
Tube A and contents after 2 minutes	114.62
Tube A and contents after 4 minutes	114.38
Tube A and contents after 6 minutes	114.38
Tube B and contents at start	120.93
Tube B and contents at end	123.38

When an oxide of copper is heated in a stream of hydrogen, the word equation for the reaction is:



- (c) Determine the mass of copper and the mass of water produced in this experiment.

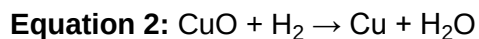
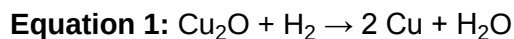
Use the table.

Mass of copper = _____ g

Mass of water = _____ g

(2)

- (d) The teacher repeated the experiment with a different sample of the oxide of copper.
- The teacher found that the oxide of copper produced 2.54 g of copper and 0.72 g of water.
- Two possible equations for the reaction are:



Determine which is the correct equation for the reaction in the teacher's experiment.

Relative atomic masses (A_r): H = 1 O = 16 Cu = 63.5

(3)
(Total 8 marks)

3.

This question is about atoms.

- (a) What does the number 19 represent in $^{19}_9\text{F}$?

(1)

- (b) How many atoms are present in one mole of fluorine atoms?

Tick (✓) **one** box.

2.03×10^{26}

☐

2.06×10^{23}

☐

6.02×10^{23}

☐

6.02×10^{26}

☐

(1)

- (c) The plum pudding model of the atom was replaced by the nuclear model.

The nuclear model was developed after the alpha particle scattering experiment.

Compare the plum pudding model with the nuclear model of the atom.

(4)

- (d) An element has three isotopes.

The table shows the mass numbers and percentage of each isotope.

	Isotope 1	Isotope 2	Isotope 3
Mass number	24	25	26
Percentage (%)	78.6	10.1	11.3

Calculate the relative atomic mass (A_r) of the element.

Give your answer to 3 significant figures.

Relative atomic mass = _____

(2)

(Total 8 marks)

4.

A student investigated the reactions of copper carbonate and copper oxide with dilute hydrochloric acid.

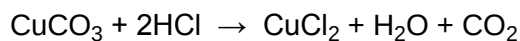
In both reactions one of the products is copper chloride.

- (a) Describe how a sample of copper chloride crystals could be made from copper carbonate and dilute hydrochloric acid.

(4)

- (b) A student wanted to make 11.0 g of copper chloride.

The equation for the reaction is:



Relative atomic masses, A_r : H = 1; C = 12; O = 16; Cl = 35.5; Cu = 63.5

Calculate the mass of copper carbonate the student should react with dilute hydrochloric acid to make 11.0 g of copper chloride.

Mass of copper carbonate = _____ g

(4)

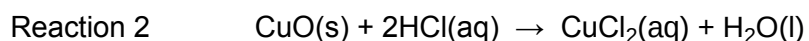
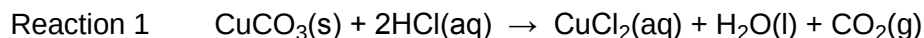
- (c) The percentage yield of copper chloride was 79.1 %.

Calculate the mass of copper chloride the student actually produced.

Actual mass of copper chloride produced = _____ g

(2)

- (d) Look at the equations for the two reactions:



Reactive formula masses: $\text{CuO} = 79.5$; $\text{HCl} = 36.5$; $\text{CuCl}_2 = 134.5$; $\text{H}_2\text{O} = 18$

The percentage atom economy for a reaction is calculated using:

$$\frac{\text{Relative formula mass of desired product from equation}}{\text{Sum of relative formula masses of all reactants from equation}} \times 100$$

Calculate the percentage atom economy for Reaction 2.

Percentage atom economy = _____ %

(3)

- (e) The atom economy for Reaction 1 is 68.45 %.

Compare the atom economies of the two reactions for making copper chloride.

Give a reason for the difference.

(1)

(Total 14 marks)

5.

A scientist produces zinc iodide (ZnI_2).

This is the method used.

1. Weigh 0.500 g of iodine.
2. Dissolve the iodine in ethanol.
3. Add an excess of zinc.
4. Stir the mixture until there is no further change.
5. Filter off the excess zinc.
6. Evaporate off the ethanol.

- (a) Ethanol is flammable.

Suggest how the scientist could carry out **Step 6** safely.

(1)

- (b) Explain why the scientist adds excess zinc rather than excess iodine.

(3)

- (c) Calculate the minimum mass of zinc that needs to be added to 0.500 g of iodine so that the iodine fully reacts.

The equation for the reaction is:



Relative atomic masses (M_r): Zn = 65 I = 127

Minimum mass of zinc = _____ g

(3)

A different scientist makes zinc iodide by the same method.

The scientist obtains 12.5 g of zinc iodide.

The percentage yield in this reaction is 92.0%.

- (d) What is the maximum theoretical mass of zinc iodide produced in this reaction?

Maximum theoretical mass = _____ g

(3)

- (e) Suggest **one** reason why the percentage yield in this reaction is **not** 100%.

(1)

- (f) The scientist makes a solution of zinc iodide with a concentration of 0.100 mol / dm^3

Calculate the mass of zinc iodide (ZnI_2) required to make 250 cm^3 of this solution.

Relative atomic masses (A_r): Zn = 65 I = 127

Mass = _____ g

(3)

(Total 14 marks)

Mark schemes

1.

(a) liquid gas

1

(b) (boiling point) increases (down the table / group)

1

(because) the relative formula / molecular mass increases

or

(because) the size of the molecule increases

1

(so) the intermolecular forces increase (in strength)

allow (so) the forces between molecules increase (in strength)

1

(so) more energy is needed to overcome the intermolecular forces

allow (so) more energy is needed to separate the molecules

*do **not** accept a reference to breaking bonds unless specifically between molecules*

1

(c) boiling point is a bulk property

allow boiling point is related to intermolecular forces (so more than one molecule is involved)

1

(d) the gas / halogen is toxic

allow the gas / halogen is poisonous / harmful allow to prevent inhalation of the gas / halogen

ignore deadly / lethal

1

(e) (going down the group) the outer electrons / shell become further from the nucleus

allow energy level for shell throughout

allow the atoms become larger

allow the number of shells increases

ignore the number of outer shells increases

1

(so) the nucleus has less attraction for the outer electrons / shell

allow (so) the nucleus has less attraction for the incoming electron

allow (so) increased shielding between the nucleus and the outer electrons / shell

allow (so) increased shielding between the nucleus and the incoming electron

1

(so) an electron is gained less easily

1

(f) 4.48 (g iron) **and** 8.52 (g chlorine)

1

(moles Fe = $\frac{4.48}{56}$ => 0.08

allow correct calculation using incorrectly calculated mass of iron

1

(moles Cl = $\frac{8.52}{35.5}$ => 0.24

allow correct calculation using incorrectly calculated mass of chlorine

allow (moles Cl₂ = $\frac{8.52}{71}$ => 0.12

1

(Fe : Cl = 0.08 : 0.24 => 1 : 3

allow correct calculation using incorrectly calculated moles of iron and / or chlorine

2 Fe + 3 Cl₂ → 2 FeCl₃

allow multiples / fractions

allow a correctly balanced equation including Fe and Cl₂ from an incorrect ratio of Fe : Cl

*allow 1 mark for Fe **and** Cl₂ (reactants) **and** FeCl₃ (product)*

or

*allow 1 mark for Fe **and** Cl₂ (reactants) **and** a formula for iron chloride correctly derived from an incorrect ratio of Fe : Cl (product)*

2

[16]

2.

(a) to make sure all of the oxide (of copper) has reacted

or

to make sure all water (produced) is removed

ignore to ensure complete reaction unqualified

ignore to make sure all of the hydrogen has reacted

1

(b) to prevent hydrogen escaping (into the air)

1

(because) hydrogen is explosive

ignore hydrogen is flammable

1

(c) (mass of copper) 8.66 (g)

1

(mass of water) 2.45 (g)

1

(d) moles Cu = 0.04

or

$$\frac{2.54}{63.5} = 0.04$$

1

moles H₂O = 0.04

or

$$\frac{0.72}{18} = 0.04$$

1

ratio = 1:1 so equation 2 is correct

1

alternative approach A

(calculating mass of water from copper)

$$\text{moles Cu} = 0.04 \text{ or } \frac{2.54}{63.5} = 0.04(1)$$

$$0.02 \times 18 = 0.36 \text{ (g of water for equation 1) (1)}$$

$$0.04 \times 18 = 0.72 \text{ (g of water) so equation 2 is correct (1)}$$

alternative approach B

calculating mass of copper from water)

$$\text{moles H}_2\text{O} = 0.04 \text{ or } \frac{0.72}{18} = 0.04 (1)$$

$$0.08 \times 63.5 = 5.08 \text{ (g of copper for equation 1) (1)}$$

$$0.04 \times 63.5 = 2.54 \text{ (g of copper) so equation 2 is correct (1)}$$

alternative approach C

(mass ratio)

(copper : water for equation 1)

$$127 : 18 = 7.06 : 1 (1)$$

(copper : water for equation 2)

$$63.5 : 18 = 3.53 : 1 (1)$$

$$2.54 : 0.72 = 3.53 : 1 = 63.5 : 18$$

so equation 2 is correct (1)

3.

- (a) mass number

allow the number of protons + neutrons

1

- (b) 6.02×10^{23}

1

- (c) **Level 2 (3-4 marks):**

Scientifically relevant features are identified; the ways in which they are similar / different is made clear.

Level 1 (1-2 marks):

Relevant features are identified and differences noted.

Level 0

No relevant content.

Indicative content

similarities

- both have positive charges
- both have (negative) electrons
- neither has neutrons

differences

plum pudding model	nuclear model
ball of positive charge (spread throughout)	positive charge concentrated at the centre
electrons spread throughout (embedded in the ball of positive charge)	electrons outside the nucleus
no empty space in the atom	most of the atom is empty space
mass spread throughout	mass concentrated at the centre

4

- (d)
$$\frac{(24 \times 78.6) + (25 \times 0.1) + (26 \times 11.3)}{100}$$

or

$$(24 \times 0.786) + (25 \times 0.101) + (26 \times 0.113)$$

$$= 24.3$$

1

1

an answer of 24.3 scores 2 marks

[8]

4.

- (a) add excess copper carbonate (to dilute hydrochloric acid)
accept alternatives to excess, such as 'until no more reacts'

1

filter (to remove excess copper carbonate)
reject heat until dry

1

heat filtrate to evaporate some water **or** heat to point of crystallisation
accept leave to evaporate or leave in evaporating basin

1

leave to cool (so crystals form)
until crystals form

1

must be in correct order to gain 4 marks

- (b) $M_r \text{ CuCl}_2 = 134.5$
correct answer scores 4 marks

1

moles copper chloride = (mass / M_r = 11 / 134.5) = 0.0817843866

1

$M_r \text{ CuCO}_3 = 123.5$

1

Mass CuCO_3 (=moles $\times M_2$ = 0.08178 $\times$ 123.5) = 10.1(00)

1

accept 10.1 with no working shown for 4 marks

- (c) $\frac{79.1 \times 11.0}{100}$

or

11.0×0.791

1

8.70 (g)

1

accept 8.70(g) with no working shown for 2 marks

- (d) Total mass of reactants = 152.5

1

134.5

152.5

allow ecf from step 1

1

88.20 (%)

1

allow 88.20 with no working shown for 3 marks

- (e) atom economy using carbonate lower because an additional product is made **or** carbon dioxide is made as well

allow ecf

1

[14]

5.

- (a) heat with a water bath

or

heat with an electric heater

or

allow to evaporate / crystallise at room temperature

1

- (b) to make sure that all the iodine reacts

allow so can see the reaction is complete

1

(as) excess iodine would remain in solution

1

(so) iodine could not be filtered off

allow (whereas) excess zinc could be filtered off

or

(so) the zinc iodide would not be pure

allow (so) would have to separate iodine from zinc iodide

1

(c) $\text{moles } I_2 = \frac{0.5(00)}{254} = (0.00197)$

allow moles $I_2 = 0.00197$

allow 65 g Zn: 254 g I_2

1

mass Zn = 0.00197×65 (g)

1

mass = 0.128 (g)

1

allow an expression $\frac{0.5(00) \times 65}{254}$ (g) for the first 2 marks

(d) $92.0 = \frac{12.5}{\text{maximum mass}} \times 100$

1

(maximum mass) $= \frac{100}{92.0} \times 12.5$

1

= 13.6 (g)

allow 13.5869... (g)

1

- (e) some product lost on separation
allow incomplete reaction

1

- (f) $M_r \text{ZnI}_2 = 319$

1

moles needed

$$\left(= 0.1 \times \frac{250}{1000} \right) = 0.025$$

or

mass per $\text{dm}^3 = 31.9 \text{ (g)}$

1

(mass) = 7.98 (g)

allow 7.975 / 8.0 (g)

1

an answer of 7.975, 7.98 or 8.0 (g) scores 3 marks

[14]

Examiner reports

1.

- (a) The majority scored the mark, but many thought bromine is a solid at 0 °C.
- (b) This question was not well answered. Most students stated that the boiling point increased going down the group, although some were evidently confused by the negative values in the table and stated the opposite trend. However, many students then confused boiling point with reactivity and tried to explain the trend using forces within the atom rather than forces between molecules. Students should refer to intermolecular forces rather than intermolecular bonds, to avoid any confusion about which 'bonds' they are referring to in terms of breaking them to make the substance change state.
- (c) This was very poorly answered. The idea of a bulk property was very rarely seen. There was considerable confusion between the terms atom and molecule, with many stating that bromine was diatomic so there are always two molecules. The best answers referred to the fact that boiling point relates to intermolecular forces, which cannot exist if there is only one molecule.
- (d) Many students realised that the fume cupboard would prevent inhalation of the gas, which is toxic. However, some students regard a fume cupboard as insulation to prevent the escape of heat energy, or thought it was to trap and retain the gas.
- (e) Most students knew and were able to express, that the atoms get larger going down the group. Some, however, referred incorrectly to the number of outer shells increasing. Fewer were able to express that there was a reducing attraction between the nucleus and the outer electrons or the incoming electron, with many vague references to shielding without mentioning the nucleus or the relevant electrons. Even fewer were able to express the increase in difficulty of gaining (rather than merely attracting) an electron.
- (f) Very few students scored full marks. Most students were able to calculate the mass of iron, but some calculated the mass of iron chloride instead of chlorine. They sometimes then assumed a formula of iron chloride and tried to work backwards. Of those who did calculate the masses of both iron and chlorine, most were then able to calculate the number of moles of each, but many then failed to find the simplest ratio as 1:3, instead leaving it as 0.08:0.24. Some calculated the number of moles of chlorine molecules instead of atoms; these students almost always failed to multiply by 2 to obtain the number of moles of atoms, giving a ratio of 2:3 instead of 1:3. Very few were then able to give a balanced equation; most did not use their ratio to give the formula of iron chloride as FeCl_3 . It was rare to see the formula of chlorine written correctly as Cl_2 , and iron was sometimes seen as Fe_2 .

2.

- (a) Only approximately a quarter of students scored this mark. They usually did so by identifying the need to ensure that the oxide of copper had completely reacted.

Common responses seen that were not credited included:

- to ensure complete reaction (unqualified)
- to ensure reliable / valid / accurate results.

- (b) Students found this question difficult. Approximately 80% of students scored 0 marks.

The idea that hydrogen is explosive was the most common correct response.

A common misconception was that, unless burned off, the hydrogen would affect the masses of the tubes and contents.

Common responses seen that were not credited included:

- Hydrogen is flammable.
- Hydrogen would react with the drying agent.

- (c) Students found the processing of the data quite difficult. Approximately a third of students scored both marks with a similar proportion scoring 1 mark.

Common responses seen that were not credited included:

- 1.09 g (which represents the mass of oxygen)
- 2.45 g (the mass of water) recorded as the mass of copper.

- (d) Students found this question difficult. Only approximately a quarter of students scored full marks with approximately a half scoring 0 marks.

For those who scored marks, it was usually by the main approach in the mark scheme. Here, the mark for linking the mole ratio to equation 2 was the least frequently scored.

All approaches in the mark scheme were represented and students scored best when their working was set out clearly.

The most common incorrect approach was to calculate the M_r of all the reactants and products in both equations and then simply refer to conservation of mass.